

Program 5

Demonstrate CI Integration by Running Jenkins in a Docker Container

DevOps

- program-5
 - Dockerfile
 - docker-compose.yml

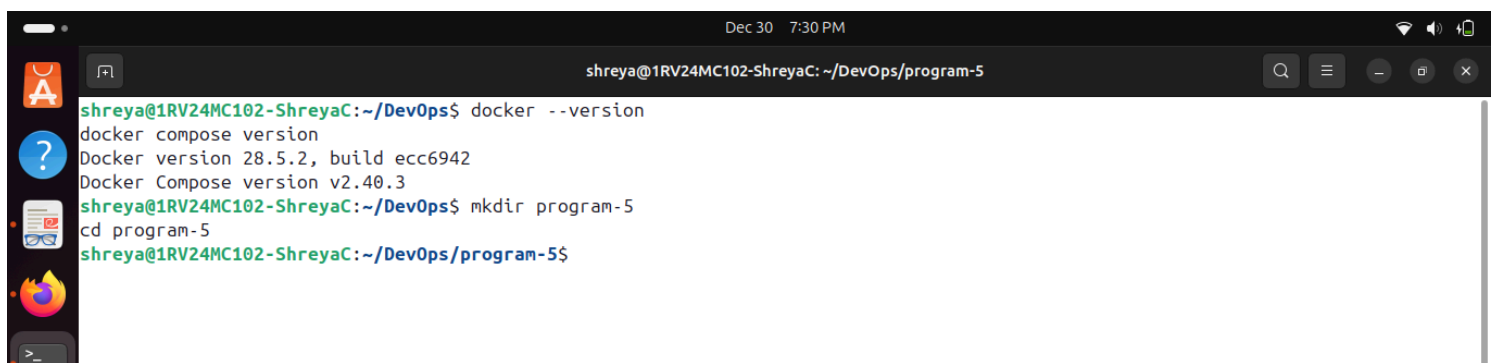
Step 1 :

docker --version – Checks whether Docker is installed and shows its version.

docker compose version – Verifies that Docker Compose is installed and working.

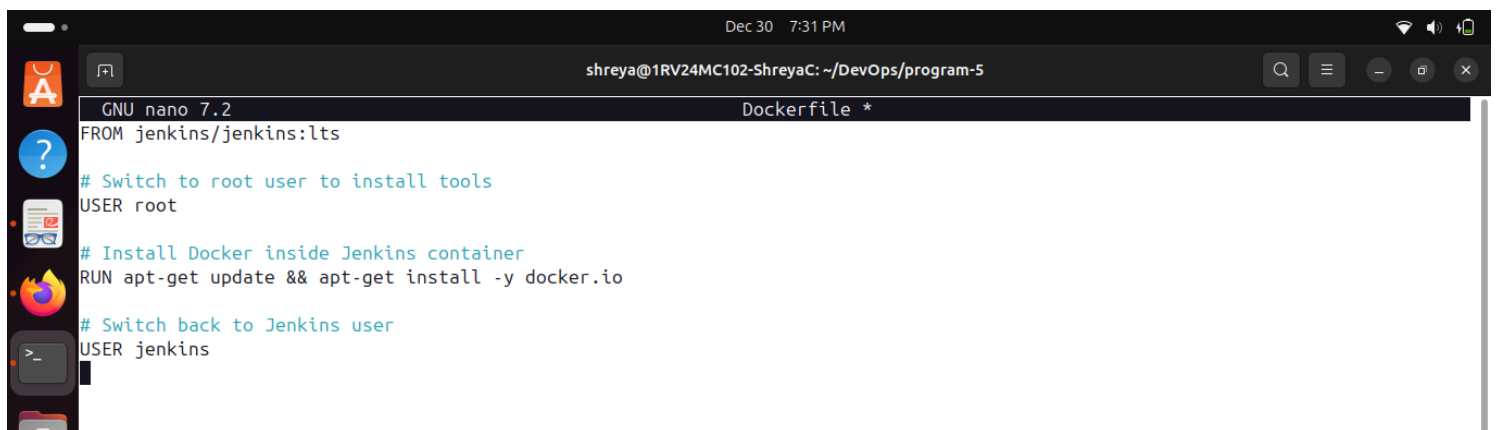
mkdir program-5 – Creates a new directory named program-5 for the project.

cd program-5 – Moves into the program-5 directory to continue the work.



```
Dec 30 7:30 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/program-5
shreya@1RV24MC102-ShreyaC:~/DevOps$ docker --version
docker compose version
Docker version 28.5.2, build ecc6942
Docker Compose version v2.40.3
shreya@1RV24MC102-ShreyaC:~/DevOps$ mkdir program-5
cd program-5
shreya@1RV24MC102-ShreyaC:~/DevOps/program-5$
```

Step 2 : Add the following code in Dockerfile



```
Dec 30 7:31 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/program-5
GNU nano 7.2 Dockerfile *
FROM jenkins/jenkins:lts

# Switch to root user to install tools
USER root

# Install Docker inside Jenkins container
RUN apt-get update && apt-get install -y docker.io

# Switch back to Jenkins user
USER jenkins
```

Step 3 : Add the following code in docker-compose.yml file

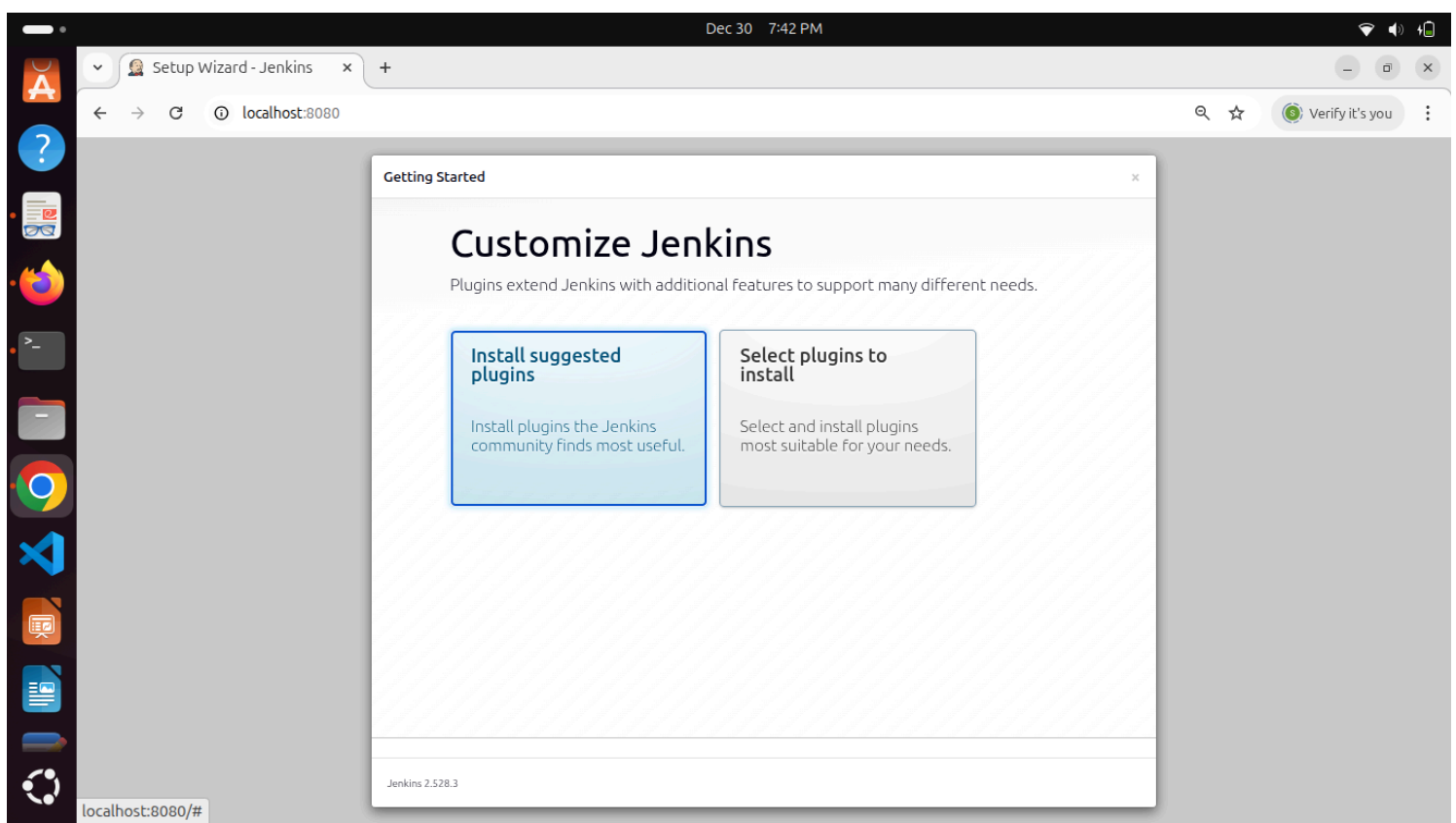
```
Dec 30 7:32 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/program-5
GNU nano 7.2 docker-compose.yml *
version: '3.8'

services:
  jenkins:
    build: .
    container_name: jenkins-ci
    ports:
      - "8080:8080"
      - "50000:50000"
    volumes:
      - jenkins_home:/var/jenkins_home
      - /var/run/docker.sock:/var/run/docker.sock

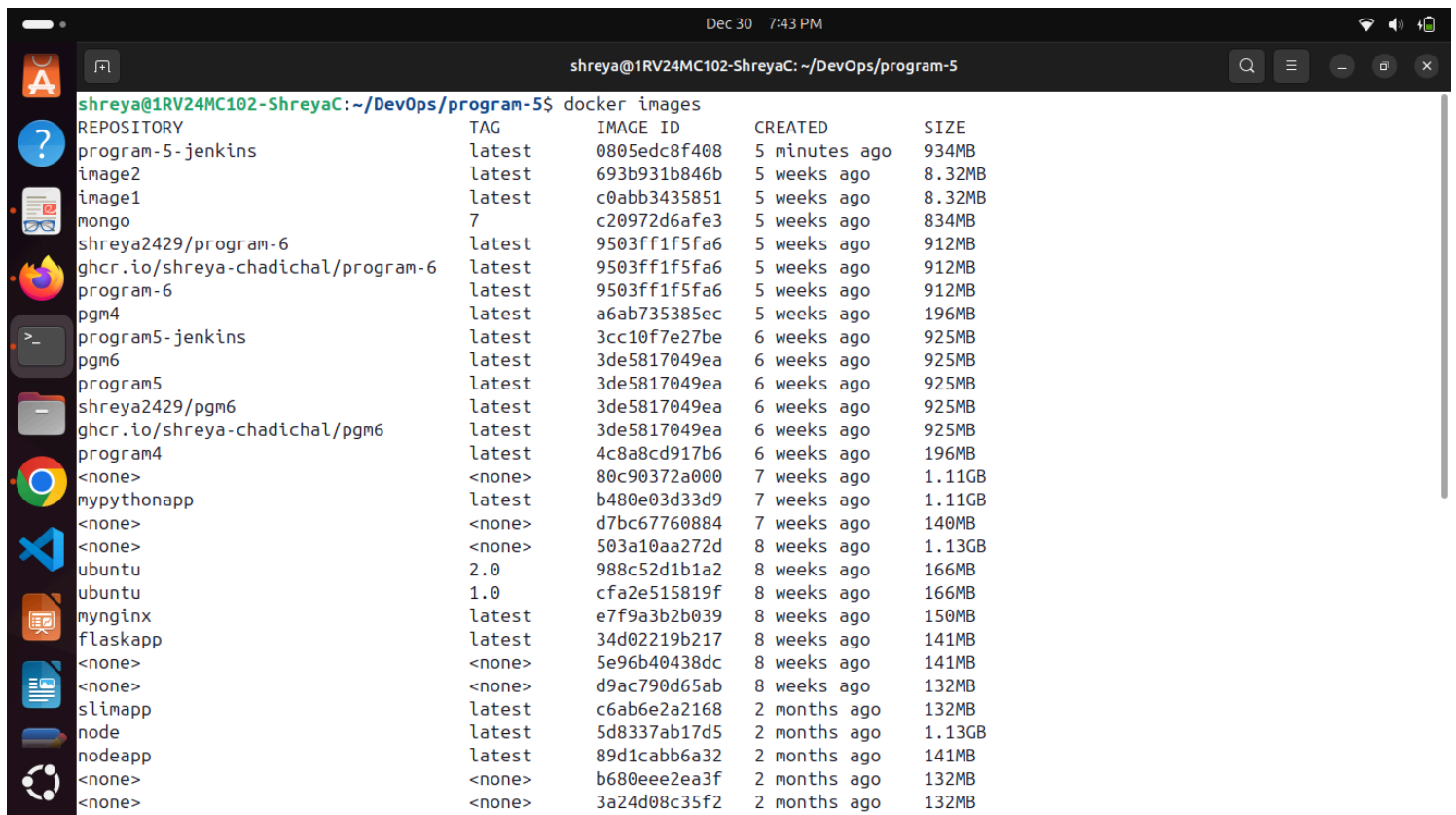
volumes:
  jenkins_home:
```

Step 4 : Build and Run the Jenkins

```
Dec 30 7:41 PM
shreya@1RV24MC102-ShreyaC: ~/DevOps/program-5
jenkins-ci | 2025-12-30 14:11:31.185+0000 [id=56] INFO jenkins.InitReactorRunner$1#onAttained: Configuration for all jobs updated
jenkins-ci | 2025-12-30 14:11:31.213+0000 [id=75] INFO hudson.util.Retrier#start: Attempt #1 to do the action check updates server
jenkins-ci | 2025-12-30 14:11:31.855+0000 [id=51] INFO jenkins.install.SetupWizard#init:
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> Jenkins initial setup is required. An admin user has been created and a password generated.
jenkins-ci | [LF]> Please use the following password to proceed to installation:
jenkins-ci | [LF]> ec8a295db07543e795b444acaeaf9b58
jenkins-ci | [LF]> This may also be found at: /var/jenkins_home/secrets/initialAdminPassword
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | [LF]> *****
jenkins-ci | 2025-12-30 14:11:40.046+0000 [id=51] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
jenkins-ci | 2025-12-30 14:11:40.073+0000 [id=39] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
jenkins-ci | 2025-12-30 14:11:41.668+0000 [id=75] INFO h.m.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
jenkins-ci | 2025-12-30 14:11:41.669+0000 [id=75] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
w Enable Watch
```



Step 5: List the docker images and delete the docker image





```
shreya@1RV24MC102-ShreyaC:~/DevOps/program-5$ docker image rm program-5-jenkins:latest
Untagged: program-5-jenkins:latest
Deleted: sha256:0805edc8f408b440768350f97dcca04a0d9da3f712bb3ed85a098066537faef1
shreya@1RV24MC102-ShreyaC:~/DevOps/program-5$
```